Indexing – Practice Problems

DBMS / Database Systems – Numerical Problems on Primary, Secondary, Dense, Sparse and Multilevel Indexing

Problem 1. An EMPLOYEE data file contains 30,000 records, where each record occupies 100 bytes and the block size is 500 bytes. A dense index is created on the EMP_ID field, which requires 8 bytes, and each record pointer requires 6 bytes. Determine the blocking factor of the data file, number of data blocks, index entry size, index blocking factor, number of index blocks, and the number of block accesses required to search for an employee using binary search on the index. Also determine the average number of block accesses required using a linear search.

Problem 2. A STUDENT data file contains 20,000 records, where each record occupies 80 bytes and the block size is 400 bytes. The file is sorted on the Student_ID field, which requires 6 bytes. A sparse primary index is created on Student_ID, and each block pointer requires 5 bytes. Determine the blocking factor of the data file, number of data blocks, index entry size, index blocking factor, number of index blocks, and the maximum number of block accesses required to search for a student using binary search on the index. Compare the result with the average number of block accesses required for a linear search.

Problem 3. A CUSTOMER data file contains 50,000 records, where each record occupies 120 bytes and the block size is 600 bytes. An index is created on the Customer_ID field, which requires 10 bytes, and each record pointer requires 6 bytes. Determine the blocking factor and number of data blocks. Calculate the number of index blocks when a dense index is used and when a sparse index is used, assuming one index entry per data block for the sparse index. Compare the storage requirements and search efficiency of the two indexes.

Problem 4. A LIBRARY data file contains 12,000 book records, where each record occupies 150 bytes and the block size is 600 bytes. The file is sorted according to Book_ID. A primary index is created on Book_ID, which requires 7 bytes, while a block pointer requires 5 bytes. Determine the blocking factor of the data file, number of data blocks, index entry size, index blocking factor, number of primary-index blocks, and the number of block accesses required to locate a particular book using binary search on the primary index.

Problem 5. An EMPLOYEE data file contains 40,000 records, where each record occupies 100 bytes and the block size is 500 bytes. The file is physically ordered according to Emp_ID, but a secondary index is created on the Department field. The Department field requires 4 bytes and each record pointer requires 6 bytes. Determine the blocking factor of the data file, number of data blocks, index entry size, index blocking factor, and number of secondary-index blocks. Assuming binary search is performed on the secondary index, determine the number of index-block accesses required to locate an employee.

Problem 6. A STUDENT data file contains 100,000 records, where each record occupies 100 bytes and the block size is 1,000 bytes. A sparse primary index is created using Student_ID as the search key. The Student_ID requires 8 bytes and each block pointer requires 6 bytes. Determine the blocking factor of the data file, number of data blocks, index entry size, index blocking factor, and number of first-level index blocks. If a second-level index is created on the first-level index, determine the number of second-level index blocks and the number of levels required to index the entire file. Finally, determine the maximum number of block accesses required to locate a student.

Problem 7. A PRODUCT data file contains 60,000 records, where each record occupies 90 bytes and the block size is 450 bytes. A dense index is created on Product_ID. The Product_ID requires 8 bytes and each record pointer requires 6 bytes. Determine the blocking factor of the data file, number of data blocks, index entry size, index blocking factor, and number of index blocks. Calculate the average number of block accesses required for a linear search and compare it with the maximum number of block accesses required using binary search on the index. Also determine the percentage reduction in block accesses achieved by using the index.

Problem 8. An ACCOUNT data file contains 48,000 records, where each record occupies 160 bytes and the block size is 640 bytes. The file is sorted on Account_No. A sparse primary index is created on Account_No, where the account number requires 10 bytes and each block pointer requires 6 bytes. Determine the blocking factor of the data file, number of data blocks, index entry size, index blocking factor, and number of index blocks. Calculate the number of block accesses required using binary search on the index, including the final data-block access. Also determine the average number of block accesses required for a linear search and compare the efficiency of the two approaches.